

CO2 Assessment -03

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Course code : ITA1408

Course Name : Ethical Hacking.

Topic 1: Footprinting Tools - Conducting competitive intelligence.

Tools used:

1. WHOIS

Purpose: To find domain registration details.

Information collected:

- Domain name
- Domain owner
- Registration date
- Expiry date
- Register details
- Name servers

Example:

Whois obctechologies.com

How it helps: WHOIS provides information about the company's domain registration, helping identify ownership & domain management.

2. theHarvester

Purpose: To collect employee email addresses and subdomains.

Information collected:

- Employee email IDs
- public subdomains
- Host names

Example:

thetannerster -d abctechnologies.com -b google.

How it helps: It gathers publicly available employee emails and subdomains that may be useful during reconnaissance.

3. BuiltWith

Purpose: To identify the website's technology stack.

Information collected:

- Webserver, CMS
- programming language, javascript libraries.

How it helps: Knowing the technologies used helps understand the company's web infrastructure.

4. Shodan

Purpose: To identify publicly exposed systems and services.

Information collected:

- public IP addresses
- open ports
- Running services

How it helps: It helps identify internet-facing systems that may require security monitoring.

Conclusion:-

Passive footprinting collects publicly available information without directly contacting the target. It helps organization understand what information is exposed & improve security.

Topic 2: DNS Zone Transfers

DNS Zone Transfer Command

using dig:

```
dig AXFR @ns1.example.com example.com
```

or using host:

```
host -l example.com ns1.example.com
```

If zone transfer is successful

The DNS records may reveal:

1. A records

Shows IP addresses of servers.

Ex: `www.example.com` → `192.168.1.10`

2. MX Records

Shows mail servers

Ex: `mail.example.com`

3. NS Records

Shows authoritative DNS servers

Ex: `ns1.example.com`

`ns2.example.com`

4. CNAME Records

Shows alias names.

Ex: `ftp.example.com` → `server.example.com`

5. Subdomains

Ex:- • mail.example.com

• vpn.example.com

• dev.example.com

• hr.example.com

Importance

If DNS zone transfer is allowed:

- Internal hosts may be exposed.
- Mail servers become visible.
- Subdomains can be discovered.
- Attackers gain valuable reconnaissance information.

To prevent this, DNS administrators should allow zone transfers only to authorized DNS servers.

Conclusion:

DNS zone transfers should be restricted because they can reveal sensitive network information.

proper configuration protects the organization infrastructure.

Topic 3: Social Engineering - Email Attack and Countermeasures.

Types of Attack

This is a phishing Attack.

Phishing

phishing is a social engineering attack where attackers send fake emails or messages to steal usernames, passwords, or other confidential information.

Countermeasures

1. Verify the sender's email address

check whether the email is from the official company domain.

2. Do not click suspicious links

Hover over the link to check the actual URL before clicking.

3. Never share login credentials

legitimate IT departments never ask for passwords through email.

4. Contact the IT department

Confirm the request using official communication channels.

5. Report the email

Report the phishing email to the organization's security team.

6. Enable Multi-Factor Authentication (MFA)

Even if a password is stolen, MFA provides an extra layer of protection.

7. Keep antivirus and software updated.

Security updates help protect against malware and phishing attacks.

How these countermeasures prevent credential theft:

- Verifying the sender prevents trusting fake emails.
- Avoiding suspicious links prevents visiting fake login pages.
- Not sharing passwords keeps accounts secure.
- Reporting phishing emails helps protect other employees.
- MFA adds extra security even if passwords are compromised.

Conclusion :-

phishing is one of the most common social engineering attacks. Employee awareness, careful verification and security measures such as MFA greatly reduce the risk of credential theft.